Design and Technology NEA

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10XO

Mr Laher

Problem 1

Delicate Plates

Since the majority of plates are made of glass, porcelain or ceramic, they make a highly unpleasant sound when scratched with a fork, they easily break and when they break, they shatter meaning that it is difficult to clean up all of the shards of plate and they shards are very dangerous as accidentally stepping on them could severely injure the person

A study by the NIH focusing on the injuries of minors has shown that 86% of lacerations are caused by minors stepped on shards of glass in 1998.

One solution to this problem is tempered glass which is a form of glass which has been heat treated to become stronger than regular glass. One problem with this solution is that the plate can still break on impact making it not ideal for a plate

To solve this problem, I could make a plate which is hard, does not shatter and has an appropriate design and still maintain a premium feel.

Potential Client: My mother

Potential Target Market: Restaurants

Problem 2

Heavy Guitars

The average weight of a guitar is around 3.6 kg and is almost 1 metre long this makes it difficult for people including many of my guitarist friends to practice in different places. Although their passion for guitar pushes them to practice whenever they can, the heavy weight and large size of the guitar makes it difficult to do so. In addition, electric guitars require a large amount of gear to get the correct sound that one looks for and often times, this gear can be expensive and heavy further restricting people from practicing in places other than their home.

As shown in the table:
Throughout the majority of the current decade, guitar sales have been on the rise and haver increased by 6.4% in the past year alone according to Fender's 2024 sales report.

Total Guitar Sales

Year	Units	% Change	$ Retail	% Change	Avg. Price
2024	2,630,950	6.4%	$1,070,244,000	7.0%	$433
2023	2,472,700	-.3%	$1,000,676,130	11.4%	$403
2022	2,489,390	-.3%	$934,971,450	11.4%	$372
2021	2,496,185	5.0%	$839,000,000	2.2%	$353
2020	2,377,310	1.5%	$820,746,000	21.0%	$361
2019	2,341,551	20.5%	$903,261,000	-1.9%	$386
2018	1,942,625	11.4%	$921,057,000	.13%	$529
2017	1,742,498	5.6%	$922,280,000	-0.1%	$529
2016	1,648,595	23.3%	$923,522,000	21.2%	$560

A solution to this is problem to make a hollow guitar body which significantly reduce the weight of the guitar. However, hollow-body guitars have a different tone from a full body guitar.

Ways I could help:
When designing the guitar I must take into account everything such as but not limited to tone, feel and comfort while still optimising its smaller size and lightweight.

Potential Clients: Shiv, Ubai, Kenda, Rocco

Potential Target Market: Guitarists

Problem 3

Robotics Education

Although movement such as FIRST have revolutionized robotics among high schoolers, organising events and competitions all around the world encouraging more teenagers to engage in robotics, it has a long way to go before reaching all corners of the earth. For example, in countries like portugal, spain and belarus, there are no initiatives which encourage robotics. This has become hindrance to many people who are actually interested in the subject but it is either too expensive or too distant to be worthwhile

- By 2030, there could be a **global shortage of over 85 million skilled workers**, many in STEM fields. **Source:** Korn Ferry – *The Global Talent Crunch report 2023*

- **Only about 40% of schools worldwide** have access to adequate STEM learning resources. **Source:** UNESCO – global STEM education reports 2024

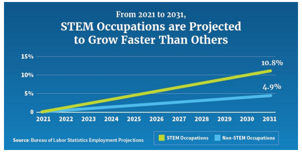
Ways I could help:

I could create a cheap robotics kit which can be easily distributed to schools all around the world and teach them about engineering and robotics.

Potential Client: Faris Tarraf

Potential Target Market: Robotics enthusiasts and young children

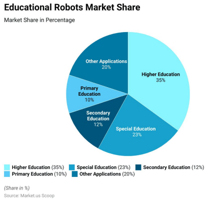
In-depth research

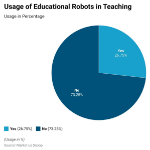


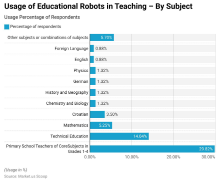
Why I chose problem 3:
Due to my background in robotics, I believe this is more suitable for my skill set and enables me to take into account more factors that I wouldn't have been able to if had picked the other 2. Furthermore, a robotics kit is much more niche and marketable compared to a guitar in which the big companies have already been established. For this reason, it would be difficult for a small company to thrive in this economic environment despite my musical background in Robotics. Additionally, I have no experience in pottery or any related field so I would most likely struggle and it would result in a less than acceptable product.

Benefits of robotics education: Robotics education is education involving the interdisciplinary application of mechanical, electrical and software engineering. However, this process of creation is not easy and only ever becomes more challenging as one advances more into the field. This difficulty comes with a silver lining: it may build perseverance within the user. Furthermore, the difficulties of building a robot often require creative workarounds to solve problems and fosters a growth mindset by forcing them through the iterative design process.. This exercises the individual's problem solving skills as described in research by Ouyang and Xu in 2024. This also helps the teenager connect disciplines together. Usually, subjects are taught separately with little to no connection between them. Robotics forces the student to connect these fields together to solve problems. These built connection have been shown to improve learning outcomes. As STEM job occupations are projected rise by 10.8% in the US alone over from 2021 to 2031 (D.O.L), STEM professionals are becoming more important now more than ever. All of these studies compound together to prove that robotics and STEM in general are becoming bigger than ever and that these students will require more help now which is the gap that I am trying to fill. This also means that I will have o focus of my design connecting numerous fields rather than specialising in a single topic.

Design implications: My product must be able to provide both a basic and a complex understanding of systems and offer both structural, electrical and programming obstacles while remaining under £40 and should target middle and high school teenagers.



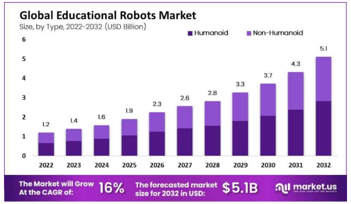




Problems with robotics education:

General market research has depicted robotics kit at an average of USD 120 - USD 150 for advanced robotics kits such as the Makerbot M2. This high cost barrier has prevented majority of students from developing their enthusiasm for robotics into a fully fledged career path.Teachers will also require training to be able to teach robotics, and this training is not vastly present in the world of today. This along with poor curriculum fit discourages teachers from integrating robotics into the classroom. This means student miss out on numerous academic advantages which would not only the students get better grades but also makes the school look more appealing. Furthermore, the advent of the internet has accelerated technological progression to astronomical amounts. Thus, maintenance along with obsolescence become regular and costly figures. To combat this, my product will be kept cheap (no more than £40) and will be reusable in different classrooms by applying a multitude of concepts within the design. The documentation for the technical concepts and the application of those concepts will be included with the robot.

Existing products: In 2023, the education robotics market had a revenue of USD 1.4 billion nad is projected to have a USD 4.3 billion revenue by 2031 as depicted by a study conducted by scoop.market. This market growth provide an opportunity for a cheaper alternative to be introduced. Furthermore, the more comprehensive a kit is, the more expensive it appears to be. As the base line is already high for robotics kits, they are usually only limited to higher educational institutions such as universities and colleges. This is reflected through their 35% market share of educational robotics as of 2026. This gap is yet to be filled by a product which can easily teach bot beginners and advanced roboticists.



Design brief

I will be designing a product that satisfies the needs and demands of middle and high school robotics enthusiasts. This product will be used over a period of 2 weeks so it must meet the requirements for that. The customer range will be between the ages of 12-18 as often they want to get started and dive deep into robotics but can only rely on small allowance money to do so. Hence, I chose an individual in this target audience. Furthermore, due to my close relationship with him, he is not afraid to tell me is a design is not aesthetically pleasing. This means I receive more critical feedback for me to develop upon.

In addition, I would like my product to be aesthetically pleasing as possible, to a meet this goal I will make the product have a clean design, making it have a sleek appeal. The finish will be matte to give it a bolder appearance and to remove any extra adhesion that a glossy colour may incur. Since tennagers often are developing their personalities, I will keep the designs simplistic.

Furthermore, the product will need to meet a range of requirements and specific needs for my client. Therefore I will need to research into my clients needs and wants through an in-depth interview and questionnaire.

The product must also be ergonomically designed and has the potential to suit multiple users, including my client and potential sellers. To do this I will do my own research on existing products, analysing the advantages and disadvantages of each product.

This products main use will be to educate teenagers on the principles of robotics for a low price so the need for it to be functional, minimalistic and cost efficient is necessary.Hence, the product will be made of reliable and cheap materials such as PLA and plywood.

The product will also be manufactured using machinery only available within the department. Therefore any manufacturing process and materials must be used efficiently so that it minimises cost and therefore maximises profit from the sales. The materials must also be available in the department meaning that every item must be used to its maximum potential to achieve the goal of making this product.

Research plan

What I plan to research	How I plan to gather the information	Why I will research it How will it help me with my project
Client information	Primary research: Interview	It will help me understand what my client wants
Aesthetics	Primary research: Interview Secondary research: Internet	It will help me design my product in an appealing way
Cost	Primary research: Interview Secondary research: Internet	It will help me set a target price on the product
Environment	Secondary research: Internet	It will help me prevent harm to the Environment
Ergonomics and anthropometric	Primary research: Client measurements	It will help me make my product comfortable to hold
Existing products	Secondary research: Internet	It will help me get ideas for my product
Size	Primary research: measurements Secondary research: Internet	It will help me set constraints on my design
Safety	Secondary research: Mr Laher and internet	It will help me prevent injury when interacting with my product
Function	Primary research: Interview Secondary research: Internet	It will help me understand what features the final robot kit should have
Materials and finishes	Primary research: Experimentation thorugh drill tests	It will help me decide which materials to use for my product
Manufacturing processes	Primary research: Experimentation through building a cube	It will help me plan how I will build my product

Have you done any robotics or electronics before?
“Yes I have.”

What was your experience like?
It was very enjoyable

What made you interested in robotics in the first place?
I first got into Robotics when I was young through a Lego Mindstorms Course my parents signed me up for.

What specifically about being in Portugal has made it hard to access robotics?
The lack of proper Robotics Teams near me, as well as the language barrier.

Have you ever tried to build something electronic or mechanical on your own?
Yes. Mechanical (I believe it would count as mechanical). A Pinewood Derby Car.

How much would you (or your parents) realistically be willing to spend on a robotics kit?
I would say 50 Euros, give or take.

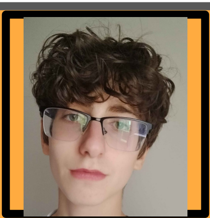
Would you prefer to follow step-by-step instructions, or figure things out yourself with hints?
I'd prefer clear instructions, instructions for individual mechanisms/features, and the ability to just figure things out myself.

Do you prefer a clean, polished look or a more barebones, mechanical look?
Both are okay, but a clean look is nicer.

Will you be using this alone or with an adult nearby?
I will be using it alone

Do you have any allergies or sensitivities to materials like certain plastics or strong-smelling paints?
Not that I know of.

Name: Faris Tarraf
Gender Male
Age: 13
Location: Braga, Portugal
Language spoken : English, Arabic
Date of birth: June 11th 2012
Handedness: Right



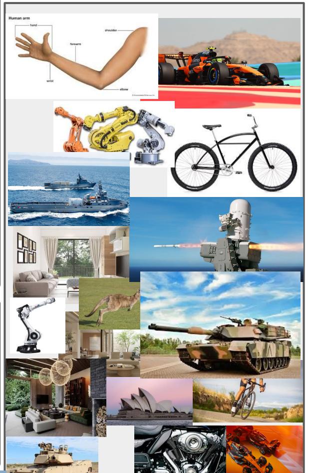
His hobbies include gaming and coding. However, due to his current relocation to portugal, he has been unable to participate in robotics. He often complains about expensive kits and the lack of programs within his area. For this reason I am making a cheap robotics kit which can be used to learn about basic engineering concepts

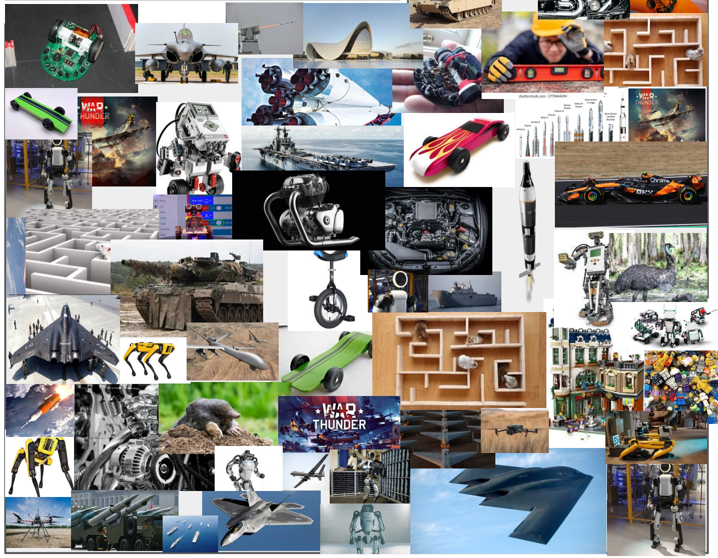
Client Needs and Wants:

Needs:
Circuit components, motors, and building materials (materials to build the chassis), cheap and fun
Wants:
External sensors such as magnetometers and cameras

Since my client’s work area is small, i will have to make sure that my parts do not take up much space

Target audience
My target audience is children and teenagers who are interested in engineering or robotics but cannot afford to do so practically.





Aesthetics

The robotics kit on the right uses blue parts which research has shown boosts creativity. This aids in the user's engagement and boost conceptualisation. Additionally, the continuous treads and the metallic coating makes the robot feel more like a tank, further interesting children and teenagers,

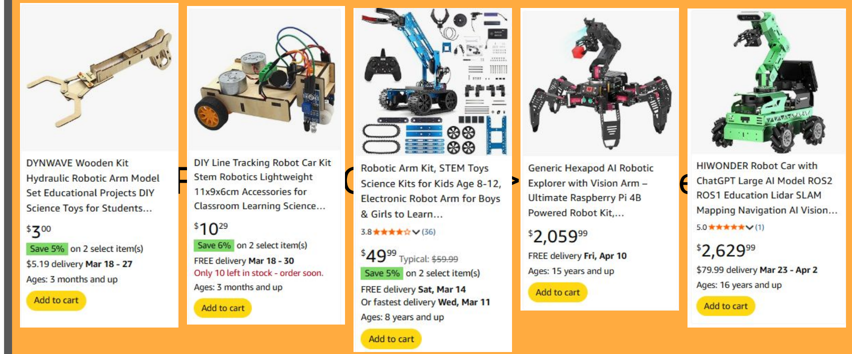
The use of the ultrasound sensors as eyes makes the robot appear more human forming a stronger connection between the user and the robot. Furthermore, .

Colour Psychology: As referenced in the first aesthetic analysis of a product, certain colours abet the user to express certain emotions. For example, blue is commonly associated with technology and engineering products .Hence, I will use blue as a main colour and red as accents;

Biomimicry: Biomimicry is the use of the natural world to solve problems that we ourselves have. Incorporating the natural world into my product, it will teach the user that a great source of inspiration for solving real world problems is to look at nature as it often has already solved the same problem but in a different context. For example, the bullet train often made large booms whenever passing through tunnel. So engineers took to the kingfisher bird for inspiration which ultimately helped with the final design We can apply the same concept to this kit. The "animal" that most people interact with are human so making a humanoid robot would build a greater connection between the robot it the user.

Design implications: From this research, I have decided to use matte and vibrant colour for my product as it seems to be backed by both scientific and market research to maximise engagement within the target audience. I have also decided to design my ideas around with real animals or real icons/ staples of pop culture

Cost



Cost of my product: The cost of the product will be determined materials I use, the current price of similar products on the market, and the time and labour that will go into making the product. Due to one of my product targets is being cheap, I will have to price the kit in the middle to low price range. The BOM on the right is an approximate of what the material costs will be. The prices listed may not be the same as what is listed on the website as prices of these components are volatile.

Part	Model	Price
MIcro controller	Arduino Uno	1.75
IMU	MPU 6050	1.21
DC Motor, Wheel	KIt	0.98
Battery holder	6 x AA Battery Case Shell	0.49
Servo Motors	MG995	15
Chassis	3D printed and laser cut	2
Fasteners	Zipties	2.2
Wires	Dupont	1.89
Motor Driver,	LN298N	0.27
Total		25.79

Design implications: Since the estimated cost of materials is USD 26/ £20. Including shipping, packaging, print time and paint, It would cost closer to £25. I will price my product at USD 50/ £35 which is severely cheaper than my competitors such as REV who is priced at USD 695 (£3700 and £512) respectively while still teaching the almost all of robotics concepts.

Environment and sustainability

Fossil fuels are fuels which were formed over millions of years from dead animals and plants. Thes often use combustion to release energy. This releases toxic gases such as sulphur dioxide and carbon dioxide into the atmosphere and contribute to global warming. Nuclear energy is energy derived from the nuclear decay of enriched uranium 235. This produces radioactive waste which is difficult to dispose of and takes centuries to decompose. Renewable energy is energy derived from renewable sources such as wind and the sun but is unpredictable and only produces energy in intervals. A finite resource is a resource which will eventually run out. An infinite resource is a resource which is able to be renews at a rate in which the new resource can replace the old resource. Though reliable versions of this does not exist, we should preserve the energy we do have. For this reason, I have decided to an use easy to manufacture material such as PLA which does not require much energy to deform but remains still at room temperature.

The recycling symbol three arrows symbolize the process of collecting, manufacturing, and purchasing products made from recycled materials. Recycling is needed to conserve natural resources like trees and minerals by using materials which were previously used to make products; reduce energy consumption and greenhouse gas emissions, decrease pollution from landfills and waste, protect ecosystems and wildlife.

LCA is a method to determine a product's impact on the environment. LCA looks at the entire life cycle of the product to prevent one impact from being shifted to another. The LCA is a standardise industrial practice which is governed by ISO 14040 and 14044.

What are the 6Rs:

1) Rethink (think about using non-recyclable materials before using them)
2) Refuse: try your best to avoid using non-recyclable materials
3) Reduce: limit the amount of non-recyclable
4) Reuse: limit the amount of materials you waste by using scrap material which otherwise would have been considered as trash
5) Repair: try fixing things before throwing them out
6) Recycle: if it is not possible to reuse the materials, try recycling the materials

Materials that can be recycled:

1) Plastics*
2) Cardboard
3) Wood
4) Metal

Materials which a cannot be recycled:

1) Styrofoam
2) Ceramic
   Teflon

* Only certain forms of this material can be recycled and only in specialised facilities

Design implications: From this research, I must use materials that are biodegradable and will have to follow ISO 14040 and ISO 14044. Hence, I am inclined to use woods, PLA and possibly metals such as aluminium. Since PLA is often recycled and can be melted down back into filament, I do not need to be overly cautious with my usage of the material during the ideation and prototyping stage. However, the ease of use of styrofoam does make it enticing for initial prototyping but the environmental harmit will cause has pushed me away from using it.

Product Analysis

Aesthetics:
The robotis mainly black with yellow and blue accents. This along with its barebones nature makes the product look aesthetically pleasing while remaining functional

Size:
The robot is a compact robot car for a tabletop. Hence can be used on desks and is easy to store and transport.

DR CLINT

Wonderful product as a STEM kit!!!

Reviewed in the United States on February 6, 2026

Verified Purchase

This Smart Robot was my introduction to Elegoo products and the experience has made me a diehard fan of Elegoo. I bought this for my 9 year old son as a gift for the holidays. This was my attempt to give him gifts would peak his interest enough to get him away from all the games and electric devices and this did the trick. At first when I first opened it I wondered if it was to much for him to assemble but boy was I worng. He enjoyed every moment of it and has not stopped using it. Its so much fun that everyone in the house is enjoying it. It was designed and packaged so well and the instuctions were very easy to follow. Love the care they took and the packaging was neat and well organized. In its default state its packed with features but the intro to Arduino IDE increases its value immensely making it possible to add more components, features and functionality. This has already paid for itself times over and its value for money is undeniable!!

Happy Shopper

so-so

Reviewed in the United States on December 28, 2025

Verified Purchase

The good: In spite of it seeming like a toy, it is a very complex electro-mechanical device. The kit is assembled with well illustrated instructions.
The so so: I don't think the average 8-12 year old could assemble without damaging or mis wiring it because of the tiny parts and no knowledge of what the component boards do. My 11 year old granddaughter who loves science, math and stem looked at it and walked away. I built it and then she took over running, programming and connecting to her phone. Then......
The bad: It started falling apart. The camera module fell off, the drive motors came loose because the fasteners all unscrewed from the turning and jerking and vibration. The smallest of nuts were lost forever. My assembly issue was to overtighten and risk breaking boards or not tight enough and have them come loose.
In conclusion: Its relatively inexpensive and does work. I think I would use Locktite on screws. Also wrenches to hold tiny nuts should be included.

Safety:
Low voltage electronics (Arduino boards and small motors). Requires supervision for younger users due to small components and wiring. It is unlikely that someones finger get stuck in any places

Cost:
The cost of the product is around USD 100 However, this is only the cost within the US. I believe this is a fair price as it offers high quality parts at a feasible price point.

Customer:
The product has clearly been designed for a customer to build themselves as it include not only the parts that are needed but also the physical tools that would be used

Environment:
The product is mostly acrylic and electronics. Hence, the product itself is not environmentally safe. However, the parts could be reused in separate projects making the product less environmentally threatening. However, the acrylic will have to be thrown away once used as it cannot be used for be used for any other project.

Function:
Allows students to **build and program robots using Arduino**.
Often includes sensors such as ultrasonic distance sensors and line followers.
Teaches coding, electronics, and robotics principles.

Product analysis

Size:
Approximately 18.3 × 4.7 × 11.2 inches.

Desktop-sized robotic arm suitable for laboratory or classroom work.

Cost:
Mid to high price for an educational robotics kit (around USD 150–USD 200).This high of a cost is due to multiple servo motors and metal construction. I believe this is unreasonable as it provides little educational value compared ot other competitors.

Safety:
Moving points such as joints and grippers are possible spaces for children to get their fingers stuck so safety precaution such as turning of the orbot must be take. However, th low voltage prevents this from causing real harm

Tom Carroll

★★★★★ **Quality Construction. Clear Assembly Videos**

Reviewed in the United States on March 2, 2018

Edition: LeArm | Verified Purchase

When I received the LeArm kit from Amazon, I was amazed at the quality of the arm, from the heavy metal base, industrial quality ball-bearing, good servos and the very clear assembly instruction videos on the Internet. It assembles easily in a few hours. I would recommend this arm over any other robot arm for robot experimenters as it will last. I intended to write about it in my monthly column in a robotics magazine and was waiting for their version of the 5-DOF hands, another manufacturer's version of which is shown in the photo. I never received those hands from LeWanSoul. I did receive their X-Arm shown to the right in the photo and found it to also be of excellent quality.

However, after a year of experimentation and demonstrations for others, I have found both of these robot arms far higher quality than any of the other robot arm kits is have tested. The LeArm uses separate 'standard' servos, each of which must be connected to a controller or Raspberry Pi or Arduino board. Depending on the position of the arm, I could manipulate almost a pound. Programming of the LeArm is fairly simple as you are dealing with standard servos that require basic pulse-width signal inputs.

The X-Arm has very similar construction but you can look at both robot arms at the Amazon site and see the physical similarities and differences in the two arms. Both use the same quality ball bearing base, though the metal structure is a bit different and the gripper/claw is identical to the other. This arm uses their LX-15D intelligent serial bus servos that can be 'daisy-chained' and the final end of the three wires of the daisy chain transfers the servo's address (each servo has a number of 1 to 6) and power through the three wires. (You can see the small connector on the end of the white cable in my photo) Your microcontroller addresses each servo independently through that cable, very similar to the Robotis Dynamixel servos. The higher cost is due to the cost of the high-cost intelligent servos.

I had a bit of trouble programming the arm with a RaspberryPi but soon got the hang of it. I ended up writing a bit of code to use it with Python, directly.

The only issue I might say is using the tiny metric screws that were all in one bag. I found some that were too long so I used washers in mounting them to the small aluminum cylinders between the servos. One of the threaded holes in the long cylinder was a few degrees off 90 degrees so I just left the screw off. The video instructions were fine though I would have preferred a written manual. I found myself 'rewinding' the video to listen again 'what the girl really said.' I also would have liked to test their two hand models as theirs seems so much better made than the other company's hands shown in my photo.

I also have their 6, 16, 24 and 32 channel servo controllers that have extensively come into use with my demonstrations. The LSC-6 controller is furnished with the LeArm and is used with the hand controller (they call it a "handle") to control the arm wirelessly. Note the receiver board in the photo of their photo of the arm.

I would highly recommend the LewanSoul robotic arms and controllers and servo testers for students from high school to advanced university instruction, as well as home experimenters. Either of their servo models are well made with metal gears and have yet to fail for me. Their web site offers much material on their products, and they have been around long enough for one to find all sorts of articles about uses of the LewanSoul robot arms and controllers.

^ Read less

Customer:
Targeted at older students, robotics hobbyists, and educators. Recommended age 14+ due to complexity and is used for learning robotics, programming, and automation.

Aesthetic:
Industrial robotic arm design made from metal. The used of CNC'ed metal gives the robot a professional appearance similar to small factory robots. Exposed mechanical joints emphasize engineering structure.

blakebart

★★★★★ **Extremely well designed and manufactured 6DoF robotic arm with remote Bluetooth to iPhone/Android connectivity**

Reviewed in the United States on June 26, 2018

Edition: LeArm | Verified Purchase

I'm surprised more people haven't reviewed this robotic arm kit. I found the kit extremely well thought out. The parts machining and anodizing are just beautiful. A very important aspect of this design is the turntable that the arm sits on. It is built using a large diameter industrial quality bearing which provides a very solid base for the arm to operate on and provide servo-controlled rotation. The servos are top quality, very smooth operation and very powerful. The gripper is pre-assembled, a nice touch, and the joint design is quite clever. LewanSoul provides excellent tutorial videos, and an extremely helpful assembly video using 3D modeling to show how things go together in sequence. Another reviewer noted the importance of sorting out all the fastener hardware, that really helped smooth the build for me. The Bluetooth connection to the free iPhone app worked great, another impressive feature of this kit, especially for the money. The PS2 controller worked fine, though I prefer the iPhone app. There is also PC interface software that is provided although I have not tried that. I'm particularly interested in connecting an Arduino to the arm and writing sketches for controlling LeArm, and Aaron at LewanSoul showed me where to find the necessary information for that. LewanSoul has been very responsive answering my questions. Oh, and check out the video for LeHand on the LewanSoul website.

46 people found this helpful

Environment:
Durable aluminum construction increases product lifespan. However, electronic components require proper disposal when damaged. Since the product can be used for a long time, it reduces the waste cause by the product.

Function:
The robot is a 6-degree-of-freedom robotic arm which can be controlled via PC, mobile app, or wireless controller. Used for pick-and-place tasks and robotics programming practice.

Product Analysis

Size:
Compact mobile robot about **90 × 180 × 130 mm** makes it well suited for classroom desks and small testing environments.

Aesthetics:
Modern educational robot design with colored metal parts and wheels. Friendly appearance suitable for classrooms and younger students. Includes a display and sensors that give the robot an interactive look.

Customer:
Designed for STEM education, schools, and coding programs and is suitable for learners aged around 12+. It is often used in robotics clubs and coding classes.

Environment:
Although it uses recyclable aluminium, it also uses ABS which can harm the environment. Since the kit uses rechargeable batteries and modular parts, nothing has to be thrown aways as it can easily be recharged, repaired or upgraded.

Safety:
Rounded edges and enclosed electronics make it safer for students as it prevents them from getting cuts or from getting shocked.

Cost:
Higher-end educational robotics kit (around USD 50–USD 250).

Higher price reflects advanced sensors, controller board, and software ecosystem.

Feature:
The kit consists of a simple 2 wheel drive which can easily be programmed in either scratch or C++. This allows the user to not only control the robot remotely but also autonomously

Materials:
Aluminum structural parts
ABS plastic casing
Electronic sensors and PCB control boards
DC motors and wheels.

SDMF1919
★★★★☆ **Very frustrating at first**
Reviewed in the United States on December 27, 2024
Size: mBot Coding Box | Verified Purchase
It was easy to build. My 10 year old did it by himself. It took us about 4 hours to figure out how to control it though. The app in the android store is only compatible with really old phones and then once you find an old phone and get it installed, it doesn't work anyway. The chrome browser app doesn't work either. Use the QR code on the mblocks website to download and install the updated android app. Then it will finally work on your current device.

Design implications:
Over all three of the products, I have noticed 2 very distinct feature: They all have tool and none of them contain batteries. Something that I had not considered when thinking about my designs is that the user may not have the correct screwdriver. Including a screwdriver in the box would remove the extra hassle on burying one. Furthermore, the lack of batteries supports my decision of not including batteries within the package. Due to both the lethality and legal nature of and form of batteries (both AA or lithium ion), It is extremely expensive to ship globally. This would defeat the original purpose of the robotics kit as a cheap and reliable resource for teaching key concepts such as detailed one such as EKF.

Scroll Saw:
The scroll saw has a sharp edge which quickly oscillates up and down to cut through wood

Advantages	Disadvantage
• Allows for curved and intricate pattern	• Not as versatile as other saws • Can only cut a single angle

Solder:
The Soldering iron has a hot tip which is used to melt solder and solidify the connection between a wire and the pins

Advantages	Disadvantage
• Can be done in under 30 seconds • The joints can withstand forces	• Creates fumes which contain lead (Depending on the solder)

Vacuum Forming:
In vacuum forming, the former heats a plastic sheet until it is soft and flexible. It then uses a vacuum to suck the plastic around a mold.

Advantages	Disadvantage
• Quick to make pieces • Simple to use	• Only useful with thermoplastics • Cannot create intricate shapes

Heat Press:
The heat press heats the textile and presses a special material onto it, forming a design on the surface.

Advantages	Disadvantage
• Easy to use • Quick to print designs • Low-Cost	• Could lead to inconsistent results

Belt Sander:
A long piece of safing paper spins around a drum, shaving off chips on the surface of the wood.

Advantages	Disadvantage
• Quick to sand parts down • Simple to use	• The surface still remains slightly rough • Cannot reach all corners of a wooden piece

Strip heater:
The wire heats up allowing it to easily cut soft materials such as styrofoam.

Advantages	Disadvantage
• Quick to make models • Good for prototypes	• Not accurate • Only works for certain materials

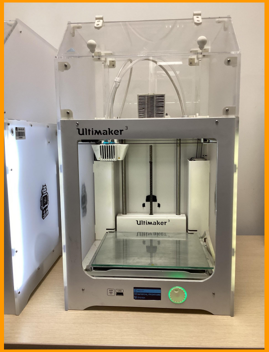
Pillar Drill:
The drill bit at the top rotates very quickly to drill through a material. The lever on the right pulls the drill straight down into the part.

Advantages	Disadvantage
• Quick • Easy to drill	• Can only drill vertically • Can be unsafe in the part is not clamped

3D Printer:

The nozzle on the top heats a plastic until it become s malleable as lays the material down onto the print bed

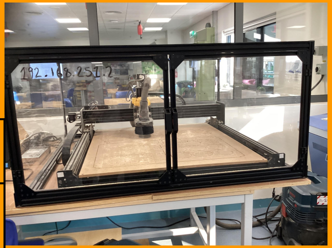
Advantages	Disadvantage
• Can make intricate pieces • Good for prototypes	• Only can make them in plastics • Slow • The print can fail



Laser Cutter:
The laser cutter burns through the wood to cuts certain pieces out of it by using a high power laser.

Advantages	Disadvantage
• Quick to make pieces • Simple to use	• Only works in 2 dimensions • Burns the material

CNC machine:
The drill bit at the top rotates very quickly. The Drill bit is moved around to remove certain parts of the material



Advantages	Disadvantage
• Good for intricate designs • Simple to use	• Slow • Only works for certain materials (depending on the drill bit • Expensive

Hand Tools:
The hand tools allow us to create designs using physical tools such as saws and handheld drills which gives us the most amount of control.

Advantages	Disadvantage
• Uses Full dexterity	• Takes time to make • Difficult to make it

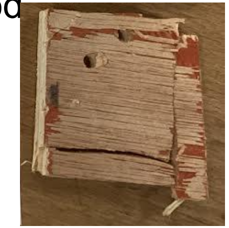
Vinyl Cutter:
The machine quickly cuts the vinyl using a very sharp blade and moves the vinyl by moving the material vertically and the blade horizontally

Advantages	Disadvantage
• Can make intricate pieces • Good for prototypes	• Only can make them in vinyl • Only works on a relatively small scale

Material	Advantages	Disadvantages	Uses
Cardboard	• Cheap • Easy to work with • Lightweight	• Difficult to differentiate between different types • Relatively weak	• Boxes • Packaging
Plywood	• Strong and durable • Resistant to warping • Lightweight • Cheap	• Susceptible to delamination • Hard to make intricate cuts • Inconsistent quality • Sometimes contain gaps	• Roof decking • Cabinets • Wall paneling
Oak	• Dense • Aesthetically pleasing grain grain • Resistant to wear • Long Lifespan	• Expensive • Heavy • Can Split • Can wear tools	• Furniture • Floors • Doors • Veneers
Pine	• Cheap • Easy to cut • Light • Aesthetically pleasing grain	• Soft • Can warp • Not moisture resistant	• Furniture • Shelves • Framing • Paneling
MDF	• Smooth • No grain • Cheap • Easy to cut	• Heavy • Poor moisture resistance • Lots of dust	• Cabinet • Mouldings • Speakers • Panels

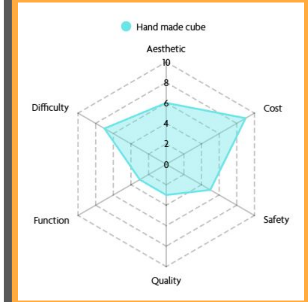
Material	Advantages	Disadvantages	Uses
Acrylic	• Clear • Light • Available in different colours • Weather resistant	• Scratches • Hard to machine • Brittle	• Signs • Screens • Lightboxes • Aquariums
PLA	• Useful for 3D printing • Biodegradable • Available in many colours	• Brittle • Not useful long term • Low strength	• Prototypes • Model Parts • Replacement Parts
Aluminium	• Lightweight metal • Resistant to corrosion • Easy to machine • Recyclable	• Soft • Expensive • Lower Strength	• Robot frames • Extrusions • Screws • Heat Sinks
Steel	• Strong • Cheap • Easy to machine • Long lasting	• Heavy • Rusts • Conductor • Hard to cut	• Cars • Appliances • Beams • Tools

Material Testing

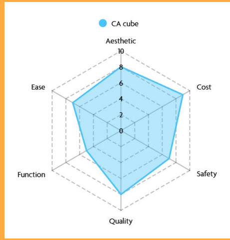
Material Name	Scratch Test	Dent Test	Drilling Test	Cutting test
Acrylic 	• Does Not scratch with a fingernail • Captured Fingerprints • Scratches with sharp objects • Scratches are only visible when shone against a light	• Brittle • Easily Dents • Quickly fractures once dented	• Quite resistant to drilling • Large amount of residue builds up around the drill hole	• Soft • Heats up quite a bit • Results in a coarse cut
MDF 	• Does not scratch	• Difficult to dent	• Easy to drill	• Quite soft • Cuts were clean
Plywood 	• Brittle • Breaks along the grain • Easily Deforms • Scratches across the grain	• Flakes • Dents Easily	• Easy to drill	• Cuts were not clean • Easy to cut

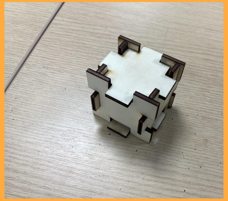
Finger Joint Cube

Making a cube with finger joints by hand was a tedious and repetitive process which did not result in a particularly good product. First, I made borders on each wood piece using the side of another wood piece. Then I marked which parts I would need to cut. 2 sides would need the all the middler sections cut out, 2 sides for need all the edge sections cut out and and the other 2 sides would need half and half cut out. To cut out the sections, I used a coping saw for coarse cuts and sandpaper for finer cuts.With the coping saw, I often cut on or beyond the marking leading to inconsistent pins and slots. As a result, I had to test each joint as many of them did not fit on the first test. Once all of the joints fit, many holes were left between ach joint which I patched up and fixed in place using wood glue. Finally, I sanded everything down to get smooth surfaces and rounded edges. Overall, this process took two lessons.



Making a cube with finger joints with CAD and CAM was drastically easier than doing it by hand. To design the parts needed for the cube, I used Techsoft. Techsoft intuitive design made the process of marking the parts a whole lot easier. Once the CAD was made, I uploaded it to a USB stick as a DXF file where I plugged it into a separate computer where, using the values written on a notebook, I calibrated the power of the laser cutter. I then uploaded my design and turned on the laser cutter. Due to the age of the school's laser cutter, I had to move the laser head out of the corner before using it. Once the cut was complete, I literally added some glue on the edges of each piece that would come in contact with another part. Then I sanded everything down ti get smooth surfaces and rounded edges. This only took a single lesson.





Design implications: From the methods of production and materials research, I have decided primarily use PLA through FDM 3D printing and maybe use laser cutting if it becomes required. This is because robotics requires very specific tolerances Which would be difficult to replicate with wood and also requires multiple iterations over 3D CAD. Overall, 3D printing and laser cutting are suitable for these roles as they can produce complex geometries easily and relatively quickly. This decision also permits modifications. Furthermore, PLA is derived from renewable resources and generates less waste during prototyping because failed parts can be reprinted without remanufacturing the entire product. Since PLA is a thermoplastic, in can be shaped in seconds by a soldering iron compared to having to use a whole other machine. Speaking of soldering irons, I will be using them to connect wires together during prototyping. Since the iterative design process with electronics requires repeatedly plugging things in and removing them, the pins will become weak. Soldering means that these connections are always strong and once they are finalised, the robot can be rigorously tested without connections becoming exposed. It also seems that metal aluminium bolts would be suitable as fasteners because they are cheap and readily available in the area where I live. Furthermore, I have decided to either plywood or cardboard for the packaging as they are both cheap materials that are easy to work with.

Size

The robot kit must be able to fit into a small box of around 300mm x 200mm 50mm with the final robot being able to easily fit on a small desk. The individual parts must be able to fix in the user's hand for easier manipulation of elements. Hence the final bot will be around 200mm x 200mm x 200mm.

Function

Problem research::

The kit must be able to accept some sort of input from a single or multiple devices and output a resulting motion or action. After completing the kit, the user must understand both basic and relatively advanced concepts of robotics

Components
A cheap microprocessor will be used as the brains of the whole robot as it is cheap and readily available. Alongside that, I will use gold MG946R servos because of their all metal gear which prevent gear skipping along with a TT motor which permits ample speed without being dangerous.

Skills it will teach: The kit will teach the use about basic concepts such as gears, belts ropes, with motors, scissor lifts along with programming skills such as sensor control, PIDF loops. These allow the user to bit only build their mechanical skills but also their coding skills

Safety

Safety Standards my product will have to pass:
CPSIA
CE Marking
BSI EN 60204-1
IEC 62368-1

The CPSIA (Consumer Product Safety Improvement Act) ensure that the product does not include any or limited harmful chemicals such as lead and Phthalates to protect children. It also mandates that the product has to go through a form of third party testing to ensure it does not harm the child.

The CE mark on products states that the product is allowed to be sold in the EU (European Union) and EEA (European Economic Area). For a product to be legally sold in the EU and the UK it must apass the EU's safety, health and environmental standards. My product must always abide by these standards as to permit sale within europe which (as stated in my in-depth research) is an extremely large market.

The BSI (British Standards Institution) is the national body which regulates technical and Safety standards across the country. It also is the body which test products for their safety and reliability and certifies them as legally sellable. They do regular tests on each product to ensure that the product still remains reliable and safe. When designing my product, I will make sure that it follows the BSI standard such as protection against electric shocks by using suitable insulation, fire hazards by using safe voltages, mechanical risks and hazards from environmental conditions use. From this, my product will receive the BSI Kitemark and this will ensure the consumer that the product is safe to use and marketable in england

Design Implications:

To keep my product safe in the hands of children I will employ a number of safety features which can be separate into 2 categories: electrical and mechanical. To prevent the children from getting shocked, I will use a battery with less than 12V, Fuses to prevent overcurrents, use proper insulation on wires, clearly label the power input and place the batteries in a safe holder. Mechanically, I will fillet the edges, make sure that the screws cannot fall out by using either threaded inserts or nuts to secure the parts, use large parts to prevent young children from choking and use safe and non toxic materials.such as PLA and cardboard.

Name	Description	Application	Advantage	Disadvantage
Spray Paint 	Spray painting is a type of painting where the paint is atomised before being released into the air and landing on the part	• Shake the spray can • Start spraying the paint while slowly moving over the part from side to side	• Fast • Smooth finish • Reaches difficult to reach places • Less physical effort: • Versatility:	• Wastes a large amount of paint • Requires preparation • Not safe • Expensive
Varnish 	A clear, protective coating made of resin, solvent, and a drying oil that provides a hard, lustrous finish to surfaces like wood	• Dip the brush into the white varnish • Varnish the part by brushing along the grain • Wait 5 min before doing another coat	• Protection • Durability • Appearance • Ease of Application	• Slow Drying • Yellowing • Difficult Cleanup • Bubbles could form • Multiple coats • Strong Odour
Wax 	Wax is a a protective coating made from natural or synthetic waxes that enhances the wood's natural grain, and offers moisture resistance	• Use a piece of cloth to rub the wax • Rub the part with the cloth	• Aesthetic • Easy To Apply • Environmentally Friendly	• Weak • Requires maintenance • Difficult to remove • Wears off quickly
Stain 	Stain is a coloured substance which 'bleeds' into wood, changing its colour while still showing the grain	• Dip the brush into the stain • Stain the part by brushing along the grain • Wait 5 min before doing another coat	• Aesthetic • Protection against moisture, mildew and sun • Easy to apply • Cheap	• Has to be replaced • Does Not hide imperfections • Few colours •
Paint 	Paint is a coloured substance which form a decorative or protective layer over a patt	• Dip the brush into the paint • Paint the part by brushing along the grain	• Aesthetic • Durable • Easy to clean • Versatile • Eco-friendly • Cheap	• Difficult to remove • Requiress prep
Lacquer	Lacquer is a clear protective layer which provides a glossy finish	• Spray the part vertically	• Fast Drying • Burable • Aesthetic	• Toxic • Yellowing

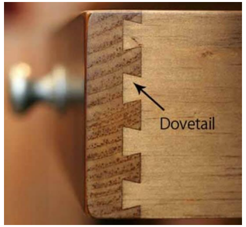
Design Implications: From this research, I have chosen to spray paint my parts as a primary background colour and to hand paint or use a stencil to draw designs onto the parts. I chose spray painting to paint large areas for its ease of use and it's variety of colours. I have chose to hand paint for intricate details as spray painting is too broad for small designs. However, I may use spray painting through a stencil for the designs if I am incapable of painting myself.I chose these painting styles over using coloured filament as There is only a select few colours to choose from and wastes a large amount of plastic when switching colors. Although I claimed to mainly use PLA for my product, I continued to research woods as it enables me variety if I am to use laser cutting to manufacture parts.

Fixing and construction methods

Wood

Finger Joint:
A finger joint cuts the wood in interlocking wedges, increasing surface area to create a stronger adhesive bond, spread out stresses and is versatile

Dovetail joint:
Instead of cutting straight wedges, flared tails' are cut out of the wood which intersect to create a strong joint. This design makes them less susceptible to lateral forces, they are more aesthetic than the finger joint and are more durable.

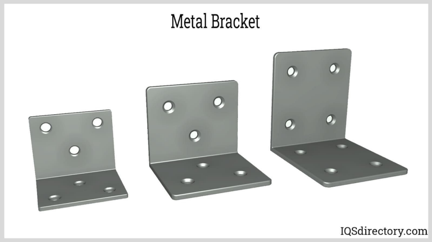


Butt joint,
a butt joint is the simplest type of wood joint making it quick to do but makes the joint weak and it may be difficult to perfectly align

Adhesive:
An adhesive is a substance which connects 2 materials together. This allows 2 different materials to be joined together differently and distributing stress and are easy to apply

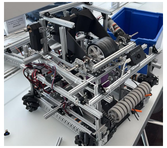
Metal

Metal Bracket:
Metal brackets are essential structural fasteners used to connect, support, and reinforce materials, commonly made from steel, aluminum, or iron. There are many types of metal bracket: L, Z, T, Flat.



Welding:
Welding works by melting base metals to physically bind the 2 pieces of metal together. This results in an extremely strong joint. However, the equipment required for this process is expensive

Rivet:
Rivets are pieces of metal which pass through holes within the metal. The end of the rivet is then deformed ,often by a hammer, creating a second head on the other side of the metal pieces. This creates a strong bond between the pieces of metal



Nuts and bolts:
Bolts are screws with a constant width. They are passed through a hole and on the other side of the pieces of metal, a nut is screw and secured tightly onto the metal

Design Implication: Since I am mainly using PLA as my material and the kit must be able to be assembled disassembled and reassembled, Nuts and bolts with brackets would be ideal because they form strong connections which can be easily broken down. Though heat press inserts can be used instead of nuts, they requie a soldering iron which would drastically increase the cost of my kit.

Ergonomics

Overview:

The ergonomics and anthropometrics of my product are important as they dictate how comfortable it is to use, controlling how likely they are to recommend it to others. More specifically, anthropometrics is the study of measuring human beings while ergonomics is how those measurements can be implemented into designs to make it more comfortable to use. To do this, I will take measurements from my client and use this anthropometric data to design a produce which is somewhat easy to build, understand and learn from. I need not only take into account o the robot physically feels when building it but also how loud it is when running, how understandable the instructions/videos are.

Percentile	5th	50th	95th	client	Source(other than client)
Hand length (mm)	155	172	189	175	PMC7720326
Hand breath (mm)	63	74	85	80	PMC7720326
Grip diameter (mm)	32	42	52	45	Springer BMC Pediatrics 2021
Hand grip strength (N)	98	178	290	130	Acta Paediatrica 2002
Finger thickness (mm)	12	15	19	16	ScienceDirect 2005/2010

Cognitive ergonomics:
This aspect of the product refers to how the product is perceived. This includes details such as the clarity of the instructions, how the code is presented and how concepts are explained. As an educational kit, this will be one of the most important aspects as without clarity, the box would be a bunch of parts.

Physical Ergonomics:
This is the aspect of the product which the user will physically interact with. Hence, the product must not consist of any sharp corners, the surfaces must be smooth. To keep the surfaces smooth, I will sand the surfaces using gradually increasing grits of sandpaper and use countersunk nuts and bolts to maintain this smoothness throughout the robot. It should also be easy to transport thus be light. The recommended carry weight for a teenager is a=10% of their body weight. Since the minimum age of the target audience is 13 y.o (whose average body weight is 45kg) the robot should weigh less than 4.5kg. In addition, the robot should also not be extremely loud. This is an important aspect of the robot as often electronic parts can make lots of noise such as coil whine (the noise which motora make while they run) which produces between 45 dB and 60 dB of noise which is significantly above the largely accepted volume of background noise as 35 dB according to the BS 8233:2014.Reducing this will make the kit hugely more bearable.

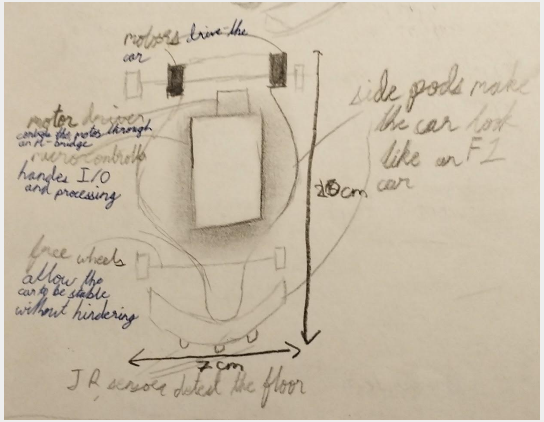
From my research, I have concluded that I must ensure that all holes must be bigger than 20mm (1mm added tolerance) to permit the user to easily access the screw holes using their fingers. Additionally, All parts other than fasteners must be larger than a 2 cm in any dimension. This ensures that the user will be able to hold the part with at least two fingers and stabilize it without falling. Furthermore, I will make sure that it weighs less than 2 kilogram. The makes sure that the user is able to easily pick up and move the parts and the full robot around with ease. The kit will also include both an instruction manual and a QR code including all of the instruction to build the kit and explain each part of it. The packaging will be a cuboid with curved bottom edges to enable easy stacky and maximum space utilisation when shipping and is comfortable to hold from the bottom.

Area	Specific Point
Aesthetics	1.1: My product should look like a unique robot but based on something real **in order to intrigue the audience for which this product is targeted for i.e it should make 75% o pupils interested in the product.** 1.2: The colours of the product will be bright and vibrant and must consist of mainly primary colours **to encourage creativity and problem solving.**
Packaging	2.1: The budget will be £30 as **it is a main priority of my product is that it is cheap to reach as many students as possible.** 2.2: I want to price my product at £40 as **it is at price low enough to be affordable but still high enough to make the product high quality.**
Client	3.1 My client is Faris Tarraf as **he already has a keen interest in robotics however, is unable to participate in competitions due to his location.** 3.2: My wider target market is children and teenagers as **they are often building their interests during this time and are often the main target for large competitions..** 3.3: Ideally, my product would be sold on amazon and would work with school to possibly integrate it into the curriculum or as a CCA **as it allows the product to reach the widest target audience.**
Environment	4.1: My product will try to reduce the amount of negative effects it will have on the environment by using much eco-friendly materials as possible as **this will make consumers more likely to buy the product**. 4.2: **Due to the use of electronics,** they product as a whole will not be recyclable. However, **the use of PLA** permits the main body to be recycled and the electrons to be reused..
Size	5.1: When built, the product will be around 20cm x 20cm by 20cm **so it can be held in the hand and can be worked on easily.**
Safety	6.1:The final product will have secure wire connections, protected electronics and double insulation. **This ensures that it is able to get the CE and Kitemark stickers**. Furthermore, it will not have sharp corners and will use less than 12V, be
Function	7.1: The kit will be able to educate the user on basic and engineering and robotics concepts such as different types of gears and lightly touching on and applying advanced concepts such as CPG gaits and Extended Kalman FIlters.
Materials	8.1: The body will be made of PLA and the electronic will be outsourced, most likely using arduino parts.. **As it is cheap and easy to prototype with and the arduino should not quickly degrade.** 8.2: A matte finish will be applied onto the body. **This is to minimise the warping of the body in high temperatures and provides a smooth , velvet-esque finish.**
User Requirements	9.1: It must be a programmable, cheap and slightly difficult to assemble as t**o encourage the user to utilise their own critical thinking to solve problems on their own and should take more than 2 hours to assemble physically**. 9.2: I will try to keep the wants integrated to a minimum **to reduce costs**.
Maintenance	10.1: The product must require minimal actions to be taken to keeps its functional value as **once its built, the user should have understood all of the concepts. Once this has happened the product has completed its mission.**
Ergonomics	11.1: The corners will be chamfered and the screws will be easily accessible so t**hat everyone, no matter their hand size, can build it..**
Manufacturing	12.1: Most of the product will be 3d printed or laser cut and secured together with Metal nuts and bolts. I may include some wooden panels as well. **This allows for more modularity of the robot and the undoing of steps.**
Product Lifespan	13.1 The product should last about a year at the least. 13.2: Once used, the 3d printed body can be thrown away and the arduino could be used to make other projects.
Packaging	14.1: The product will require compartmentalised packaging to maximise **space efficiency and to prevent parts from damaging each other.**

Kit for a line following robot

Pros:
Cheap to manufacture
Teaches basic sensors
Easy to assemble
Low part count
Lightweight
Visually engaging

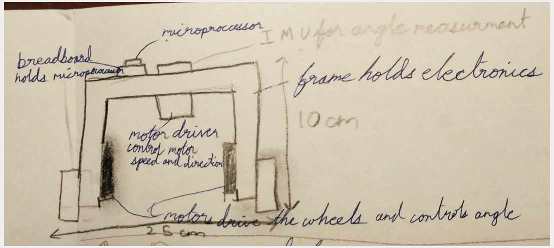
Cons:
Limited progression
Limited customisation



Kit for a 2-wheel balancing robot

Pros:
Introduces PIDF control
Teaches basic sensors and data protocols
Compact
Low material usage

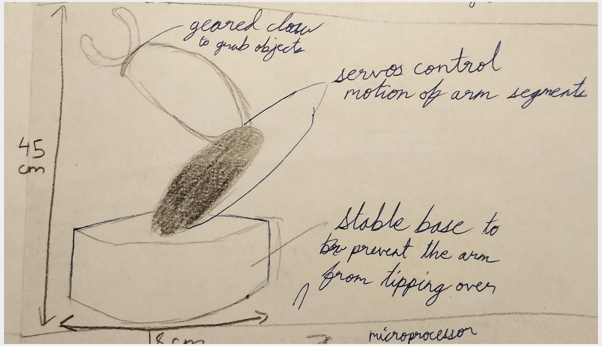
Cons:
Difficult to tune
Not visually appealing



Kit for a robotic arm

Pros:
Teaches Inverse Kinematics
Highly interactive
Easy to demonstrate
Have good mechanical aspects

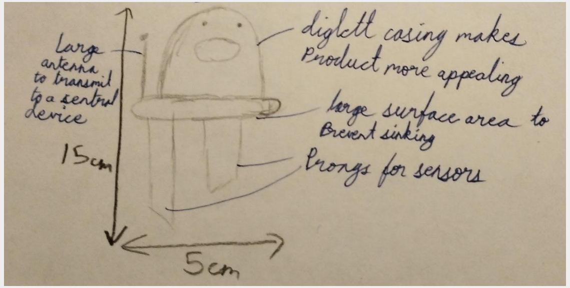
Cons:
Complex wiring may rip if code fails
Low portability
High cost



Kit for an Environmental Monitoring robot

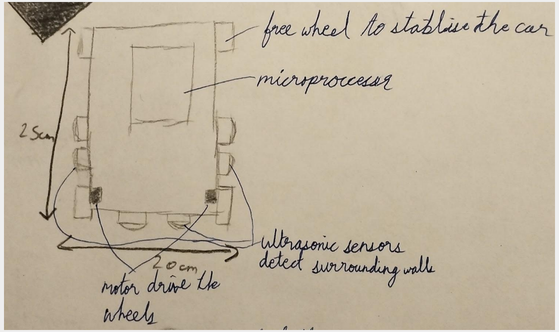
Pros:
Strong STEM links
USed in the real world
Easy to expand

Cons:
Less movement
Limited mechanical content



Kit for a maze solver robot

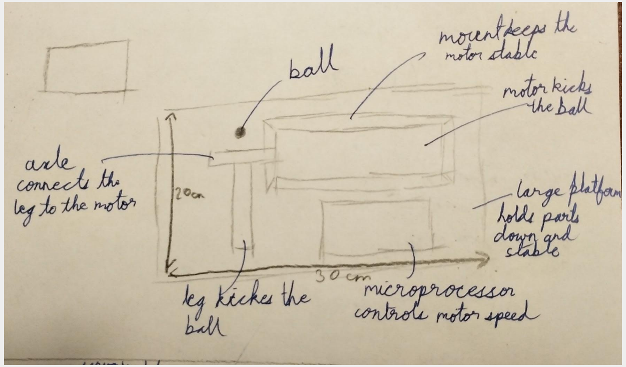
Pros:
Teaches search algorithms
Teaches sensors
Introduction to Micromouse
Can be progressed
Cons:
Difficult to debug
Requires a maze to fully test
Limited mechanical complexity



Football kicker

Pros:
Fun and engaging
Teaches motion systems
Strong visual appeal

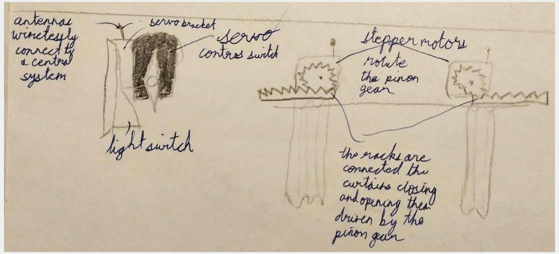
Cons:
Educational value lower
Limited expansion
Difficult to justify curriculum use



Kit for automatic light switch and curtains

Pros:
Real world application
Teaches automation
Relatively low cost

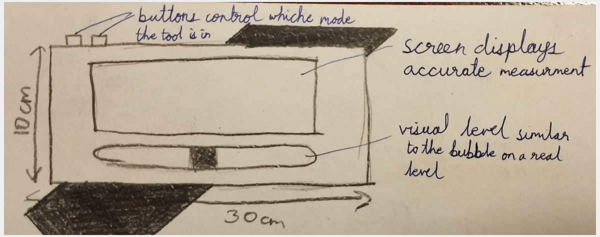
Cons:
Less engaging
Less Robotics focused
Limited mobility



Kit for an electronic leveler

Pros:
Teaches protocols and Sensors
Real life application

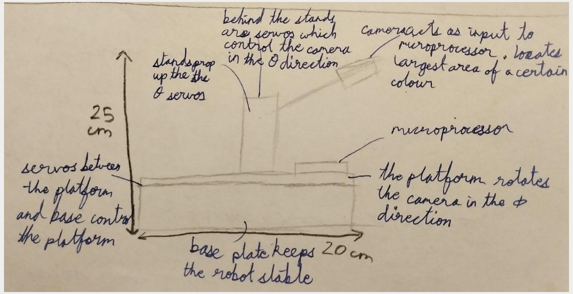
Cons:
Not visually appealing
Limited progression



Kit for a colour tracking arm

Pros:
Teaches computer vision
Visually impressive
Modular
Programming focus

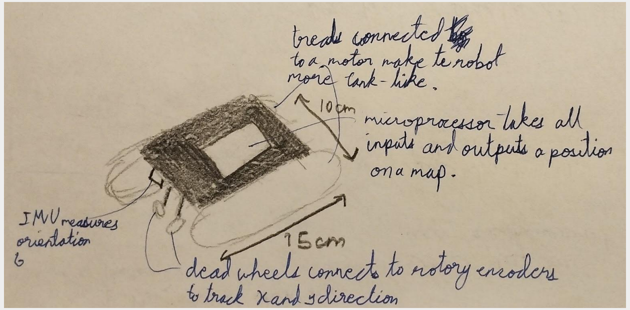
Cons:
High cost
Stationary base



Kit for a tank drive robot with odometry

Pros:
Teaches navigation
Teaches Sensors
Expandable into FTC
Real world applications

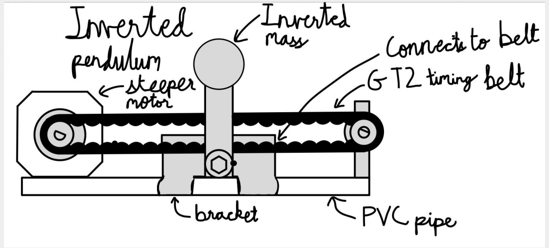
Cons:
Higher Cost



Kit for a inverted pendulum

Pros:
Teaches deep control theory
Teaches advanced mathematics

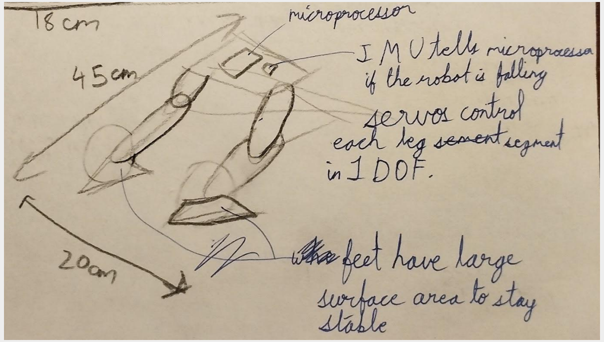
Cons:
Difficult to introduce
Easily damaged
Very Difficult to tune

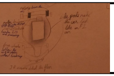
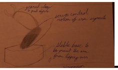
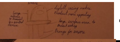
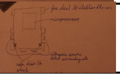
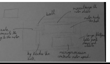
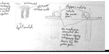
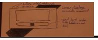
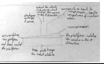
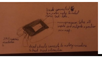
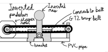
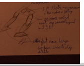


Kit for a bipedal robot

Pros:
High education potential
Teaches movement
Teaches inverse kinematics
Teaches balance
Expandable
Unique

Cons:
High cost
Long build time
Easy to fail



Idea	Aesth 1	Cost 2	Client 3	Enviro 4	Size 5	Safety 6	Func 7	Matt 8	User 9	Maint. 10	Ergo 11	Manu. 12	Life. 13	Total
 1	5	5	1	4	2	5	2	5	3	5	4	5	4	50
 2	4	5	4	4	3	4	3	5	5	4	4	5	5	55
 3	4	3	4	4	3	4	4	5	3	3	4	3	3	47
 4	5	4	4	4	4	5	2	3	2	4	4	4	4	49
 5	3	5	3	4	5	4	3	4	3	4	3	4	4	49
 6	3	5	3	4	5	4	3	4	3	4	3	4	4	49
 7	3	4	2	3	2	3	3	2	3	5	3	2	5	40
 8	5	4	2	4	2	4	2	3	2	5	3	5	5	46
 9	5	3	5	3	5	3	4	4	4	4	4	4	3	51
 10	4	5	4	3	5	5	4	3	3	4	3	2	4	49
 11	3	3	3	4	4	3	5	4	4	3	3	3	4	46
 12	5	3	4	4	5	4	5	4	5	3	4	5	5	56

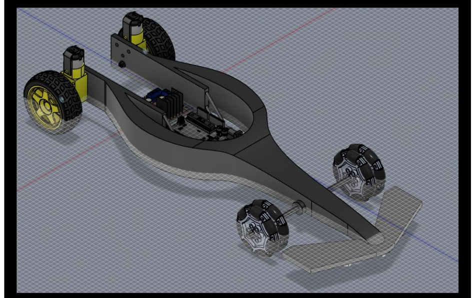
The scores were awarded between 1 and 5 through subjective thought between me and my client. Alongside, we reached out to other family members or friends on certain scores. The orange rows are the designs which will be developed designs. This matrix is a general scoring comparing to the specification and does not go in-depth into sub topics as the main purpose of this ,matix is to narrow the 12 initials ideas into 4 developed ideas

Design implications: Since Designs 1,2,9 and 10 had the 4 highest ratings, they will move onto the proposed design section.

Line following robot

The line following robot is a robot which is able to follow a black line against a white background. It uses 2 rear motors to drive the car and 2 sets of amni wheels to allow the car freedom of movement when turning. It uses a set of 3 IR sensors to detect the darkness of the ground below it. It was modeled after and F1 car and students can use it to learn about basic motor usage and how different sensors work.

The parts will come separated but would only need to be pushed together. The plastic bushings would be pushed into the nose of the robot along with a pair of omni wheels on each side. In the middle, the arduino and the motor driver will be mounted, At the back liy the motors and wheels

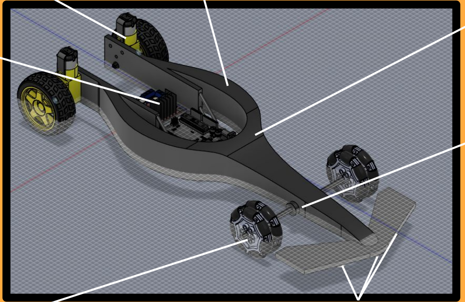


TT motors drive the back wheels

The side pods make the car look like an F1 car

The main body is PLA as it is easily 3D printable and relatively strong

The L298N motor drivers and arduino are situated in the middle to make the physics of the car more similar to F1 cars



Bushings reduce the friction between the axle and the hole.

Dual set of omni wheel ensures that at least 1 side-roller is in contact with the ground, preventing increased wheel deterioration

3 IR sensors at the bottom detect changes in the colour of the floor which informs the arduino to turn.

I would 3D print the plates using PLA. The screws, wheels and the motor will be store bought.

“I quite like this design. It is nice how it was structured around ann F1 car. This one is beautiful. It does look a bit simplistic so it would teach me as much but would offer a good entry into the field of robotics” - Faris

Pros	Cons
Cheap to produce	Takes a long time to print
Easily customisable	Omni wheels require tight tolerances
Lightweight	PLA is brittle
No fasteners required	The design is too simple to be able to be carried further

1.1	1.2	2.1	2.2	3.1	3.2	3.3	4.1	4.2	5.1	6.1	7.1	8.1	8.2	9.1	9.2	10.1	11.1	12.1	13.1

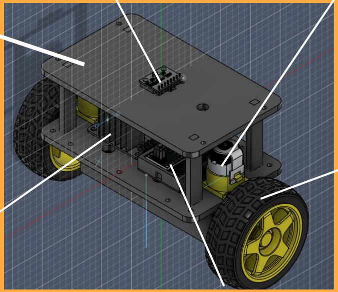
2 Wheel balancing bot

The dual wheel balancing robot is a robot which is able to self align itself upright simply through acceleration of the bottom two wheels. This will be done through the collaboration between an Inertial measurement unit and the arduino. Despise the hardware, the user will have to learn about PIDF and feedback loops to stabilize the machine. This design also allows for more further development of the mechanism as it can be adapted for other uses such as a remote controlled car or an object movement system.

The main chassis will be made of PLA as it is soft enough for threads to catch onto and is easy to 3D print cheaply.

MPU 6050 measure the rotational speed which can be converted into an absolute angle.

TT motor drive the body of the robot up by applying lateral motion different to the motion of the CoM

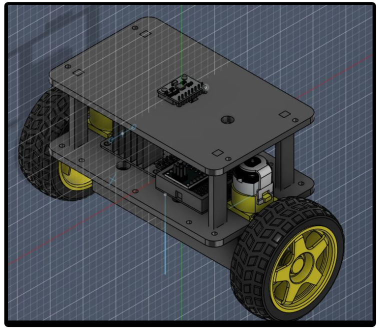


The wheel tread will be made of rubber or TPU to increase friction and stability with the floor

The L298N motor driver controls the motors direction and speed

Using an arduino nano as the micro controller as it is powerful enough for this application and its small form factor makes it ideal for this size

Assembly:
First, the cover on the breadboard would be removed and stuck into the recession on the surface of the plate. Then, the L298N and the 2x TT motors would be bolted onto the base plate. Once bolted, the standoff would be attached along with the top baser plate and MPU 6050. Finally the wheels would be attached onto the motors



I would 3D print the plates using PLA. The screws, wheels and the motor will be store bought. These will all go into a cardboard box on which I spray paint over a vinyl stencil. The related parts will be packaged together in sections made by the vacuum former.

Pros	Cons
Cheap to manufacture	Structure might break because of the tin standoffs
Low print time	Flexible PLA causes noise ito IMU readings
Teaches about algorithms such as PIDF	

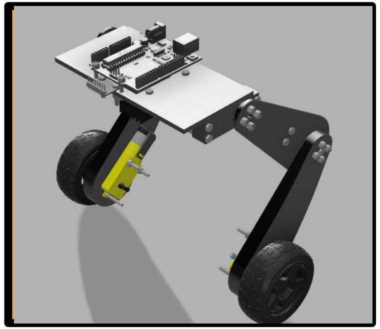
“The robot is very simplistic and appears to easily teach the basic concepts such as basic motor control, basic velocity algorithms and Sensor inputs and protocol such as UART and I2C.the aesthetic of this design is functional but lacks visual appeal for the target user group, which may reduce engagement.” - Faris

1.1	1.2	2.1	2.2	3.1	3.2	3.3	4.1	4.2	5.1	6.1	7.1	8.1	8.2	9.1	9.2	10.1	11.1	12.1	13.1

Bipedal Robot

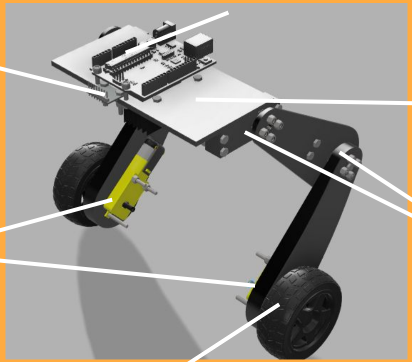
The concept for this prototype is to act as a racing robot through its freedom of movement. Due to its strong base motors, the robot will be able to sprint along straight but the individual vertical freedom on each leg allows the robot to tilts into corners. Not only this, the robot will be able to ride across terrains without being tipped over. It is through this freedom from which we can teach the users about feedback loops such as PIDF and a light introduction into kalman filters.

The user would first attach all of the servos to servo brackets. Two of these would then screw into the base plate . Then the second pair of servos brackets would mount onto the upper leg. Then, servo horns are screwed into the brackets and attached to their respective servo. Then the arduino is bolted onto the plate



An arduino uso is used as the microcontroller because it accepts a standard barrel jack connector with USB-B connectivity while maintaining an acceptable clock speed and RAM size

The MPU 6050 is used as an IMU for the robot.Since it contains both



Main chassis will be made of PLA as it is soft enough to thread into, hard enough to support electronics and easy to prototype upon

TT motors drive the robot to help it stabilize itself when still. They also help lock the wheel to increase friction when the robot walks

Tread will be manufactured out of rubber to maximise friction between the robot and the floor.

MG946R servo motors control the height of the robot and allow it to pitch up and down and roll right to left

The chassis will be made out of 3D printed PLA as it is ebay to prototype and soft and secure to thread. The fasteners, motors and wheels will be of year.

“The bipedal robot looks similar to the balancing bot but more complex. The extra servo joints make the product seem a lot more difficult which I am up for but may deter an audience who wants to get into robotics. It's also very ugly.” - Faris

Pros	Cons
Complex mechanism permit deep understanding	High print time
Soft PLA for easy threading	Screws may get stripped if used repeatedly
Free leg geometry	Takes a long time to build
	Most difficult design

1.1	1.2	2.1	2.2	3.1	3.2	3.3	4.1	4.2	5.1	6.1	7.1	8.1	8.2	9.1	9.2	10.1	11.1	12.1	13.1

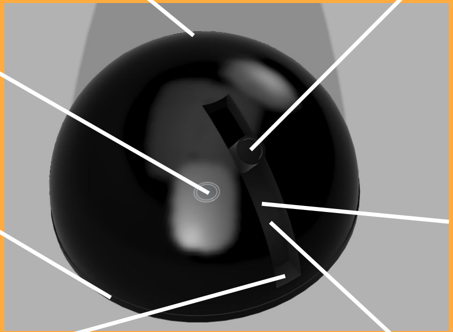
Colour tracking turret

The colour tracking turret is able to autonomously use a camera connected to an arduino to detect a certain colour and follow that colour around using a main servo controlling ϕ and a secondary servo controlling θ in a 3d polar coordinate system.

The main body will be made of PLA

MG946R gold shaft servo motors control the axis ϕ and θ of the camera

A Heavy base is used to prevent the motor from moving the base instead of the dome



Camera searches for the largest area of a certain colour in its view

All electronics will be bolted onto the platform to prevent parts from moving when they don't have to.

The whole dome rests on a lazy susan bearing. This allows the whole dome to rotate instead of just the camera

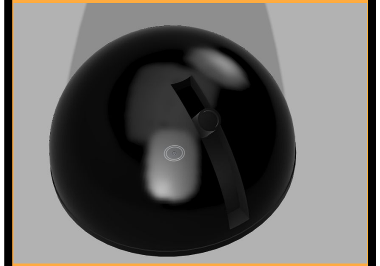
The lazy susan bearing will be made out of aluminium to present any deformations under load that may affect turing.

The kit will contain numerous parts: 4x hoods segments, 4x base segment, 2x servos,2x servo brackets, 1x arduino uno, 1x battery case 1x axle and 1x camera.

The user would start off sliding the hood segments together. Then they would attach the servos to the servo brackets. Once attached, 1 of the servo would be bolted through the bracket and into the base while the other is bolted onto of that one. Then, the camera would be attached to the axle and the axle would be attached to the second servo. This leaves the user with a physical model which will become functionable.

The overarching shell will be made with 20% infill PLA while the base will be made of 100% infill PLA. The lazy susan bearing will be store bought as it is made out of metal which is difficult to scale to larger audience when operating as a single person.

"This looks like you smashed together a missile launcher from an american warship and put Star Wars together or a planetary. Either way, I really like the turret design. However, it is a bit too static how it doesn't really move around like the other designs but overall it is a good design." - Faris

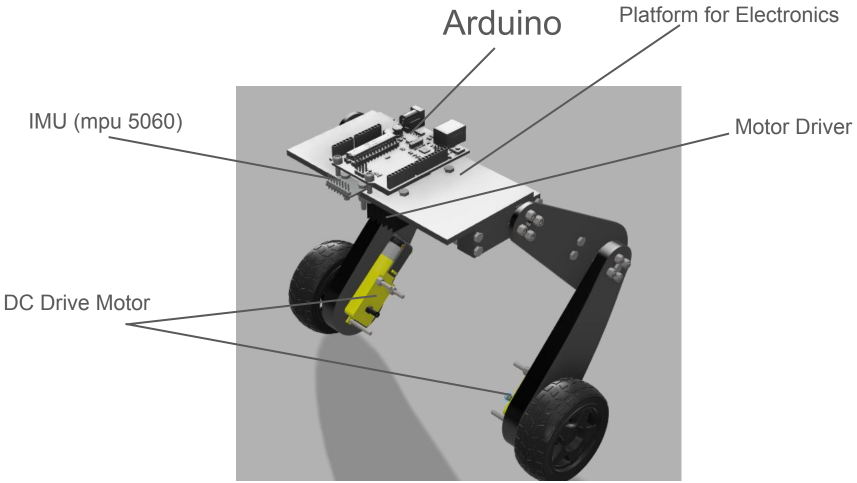


Pros	Cons
Easy to 3D print	Complex infill settings
Easy integration of lazy susan bearing	Lazy susan must be bought
PLA is suitable for static parts	PLA may warp
Low part count	Curved surface is prone to bad aesthetics

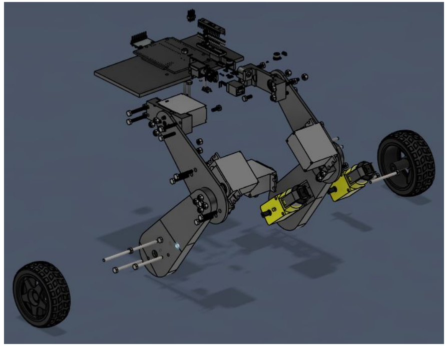
1.1	1.2	2.1	2.2	3.1	3.2	3.3	4.1	4.2	5.1	6.1	7.1	8.1	8.2	9.1	9.2	10.1	11.1	12.1	13.1

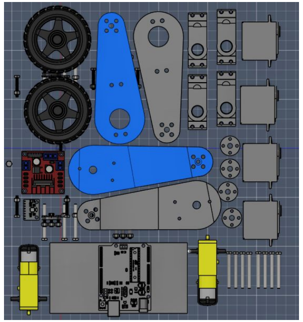
Proposed Solution

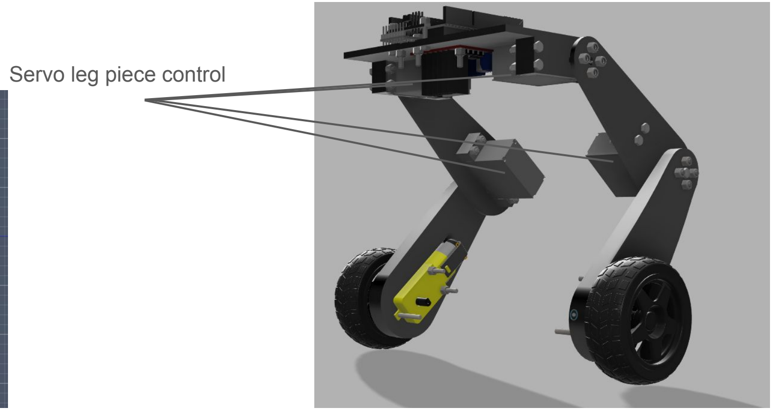
I chose the third concept design as it contains the best track for transitioning between basic and advanced robotics. It provide basic functionality with individual servo legs to advanced functionality through Bipedal CPG gaits which replicate Regular bipedal locomotion. It does hold the downside of being slightly more expensive than the other proposed solutions but I believe that the added difficulty improves the effectiveness of the design.



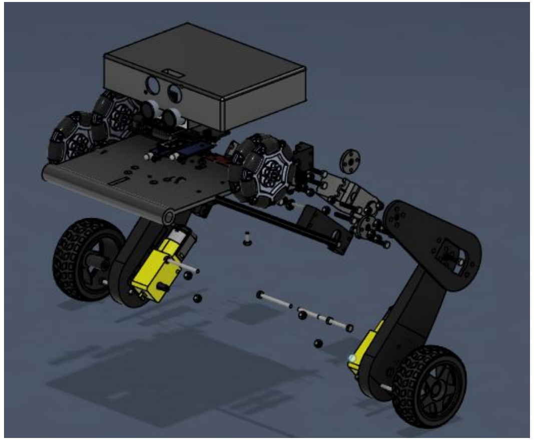
The concept for this prototype is to act as a racing robot through its freedom of movement. Due to its strong base motors, the robot will be able to sprint along straight but the individual vertical freedom on each leg allows the robot to tilts into corners. Not only this, the robot will be able to ride across terrains without being tipped over. It is through this freedom from which we can teach the users about feedback loops such as PIDF and the link between maths and engineering through inverse kinematics



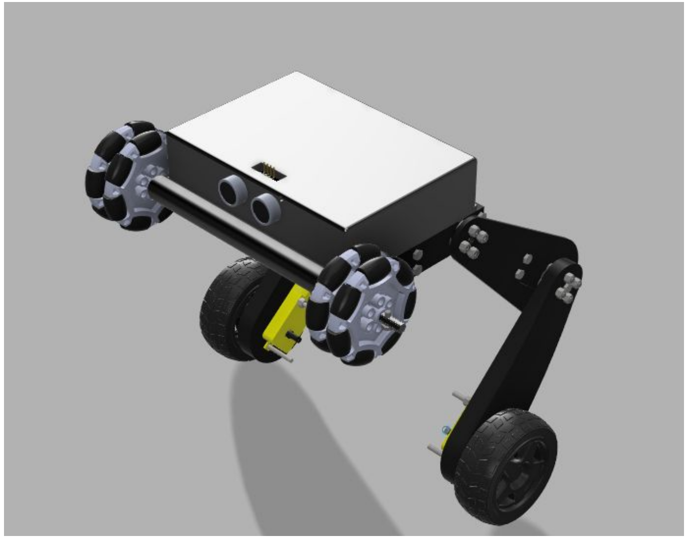


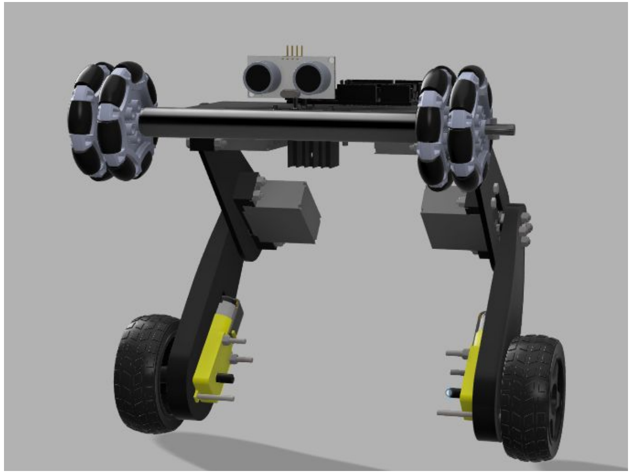


Changes from Proposed Design into V1

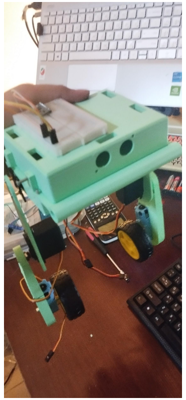


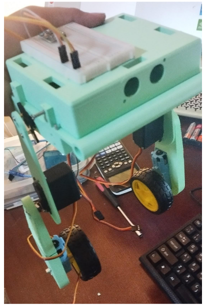
After client feedback, I had added a protective cover around the arduino and electronics. I have also removed ll sharp corners from all parts to prevent unnecessary discomfort during building. I also rearranged the location of electronics for a better weight distribution.

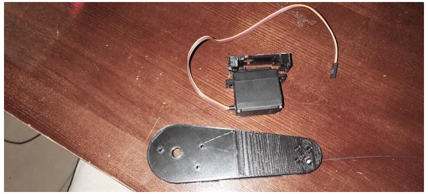


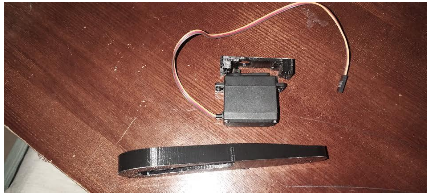


Developed V1 Physical Prototype







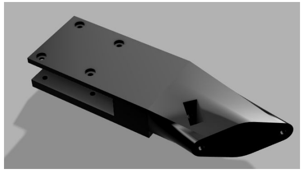


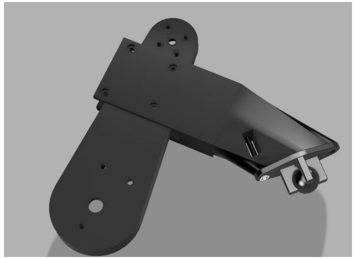
Throughout the building process, i continuously analysed the building experience and how I could improve the process. I was able to identify many flaws within my design. Often when designing, i needed the robot to stand up straight. This often proved difficult due to the bipedal nature. Although I had designed for the wires to funnel through a single hole, It proved difficult to do as wires often disconnected from the pins on the arduino. This may be because the PCB was previously used but it still presents as a possible problem. Each time I screwed something into the part, the material would often sink into the hole and the screw would extend above the surface, interfering with Electrons in sometimes even shorting the pins on the arduino uno. Furthermore, the motor brackets were often too small for the motor themselves. When 3D printing, the filament has a given thickness which interferes with part tolerances. Even though the slicer on most 3D printers accounts for this inconsistency when making the G-code, it can still present as a problem when dealing with cheaper 3D printers. This also caused issues with the fitting of the wheel have me to resort to adding it of the inside of the robot instead of the outside. It also became obvious once I started building that I did not require 2 of the servo brackets (more specifically the lower 2) as the Hole for the servo was slightly too small for he sero and the servo shaft would not extend beyond the part if the bracket was included. The Breadboard was originally designed to be placed at the top but it makes the design look more confusing and makes wring unnecessarily difficult. I had expected lateral sag within the legs as both PLA and the servo horn are flexible but the degree to which the parts flex was of concerning matter. It would also help if the cover was hinge as it would not need to tbe removed and bolted back in every time that I want to make a change.

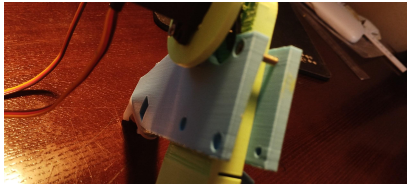
Design Implications:
From thai experience, I have decided to split the design into 2 more steps: a V1.5 and a V2. The V1.5 will include additional modules to provide basic functionality to the prototype to test electrical circuits and code. The V2 will contain all of the stated improvements of this build analysis.

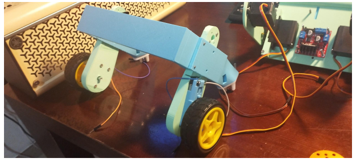
Physical Prototype V1.5

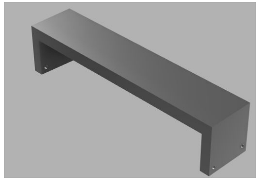
To continue to wiring and coding V1, I would require some extra pieces. As noted in my build analysis, I would require a slot to fit the batteries into. Due to voltage restrictions on the arduino uno, I would have to connect the battery pack to 2 separate sources. Furthermore, to test basic functionality I also printed 2 supporting braces on the lower legs with caster wheels on the ends. This allows the robot to easily stand upright and permits me to test basic drive functionality. Again mentioned in my build analysis, the lateral tolerance of the legs is unacceptable. To prevent a major redesign of the legs, I have opted to use a cross brace connecting the two supporting brackets.





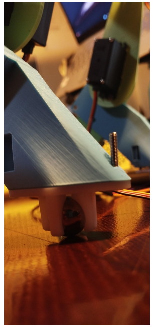


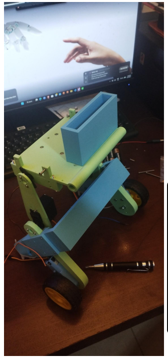


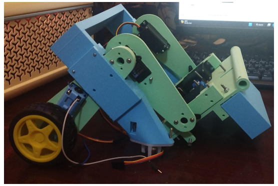


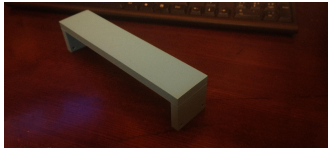
While attaching these modules onto the design, I noticed some further issues. Since the battery holder would require bolts, The main base plate would have to include holes for these. As a temporary solution, I used a soldering iron to burn a hole through the PLA. Additionally, the main cover for the electrons would not fit, hence continued with this model without the cover and one of the screws for the servo bracket won't thread fully through. These are issues to be fixed in V2. Furthermore, For a quick and easy solution on the mounting of the support brackets, I have used screws to clamp onto the leg. This will be reiterated in V2 where I will include a dedicated screw hole for the part.

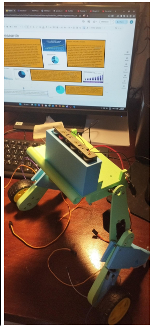
The red circle shows where the PLA seeped out when using the soldering iron to make holes.

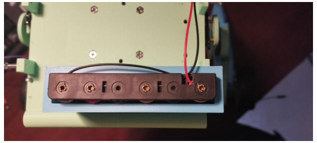


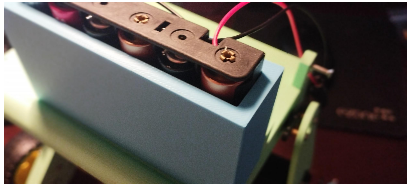


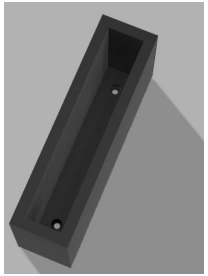


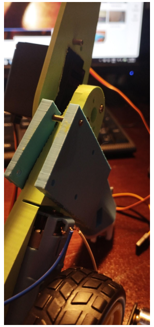












Design Implications: As a summary of the changes I will make, I will add screw holes to the base plate, I will redesign the battery holder and add the ability to attach support wheels. I will redesign the legs as to reduce lateral flex and result in inconsistent ‘foot’ placement.

Working V1.5 Prototype Continued

Developed Prototype V2

Changes made:

Add hole in Lid for USB access

Add holes for Pin access

Allow for 2 batteries

Countersink Screws

Enlarge motor brackets space for tolerance

Remove second set of servo brackets

Attach breadboard and inside the enclosure

Add a hinge for the Cover.

What I could do in the future

PCB Design

Allow for different boards to be mounted

Maybe use a custom board

Deciding between using LQR and PIDF

PIDF vs LQR vs MPC

Conclusion:
Use STM32F4 for main processing and an esp32 for communications